US012404884B2

(12) **United States Patent**
Schmidt et al.

(10) **Patent No.: US 12,404,884 B2**
(45) **Date of Patent: *Sep. 2, 2025**

(54) CYCLONIC FLOW-INDUCING PUMP

(71) Applicant: **Vortex Pipe Systems LLC**, Austin, TX (US)

(72) Inventors: **Paul Wayne Schmidt**, Carlton, OR (US); **Avijit Ghosh**, Austin, TX (US)

(73) Assignee: **Vortex Pipe Systems LLC**, Austin, TX (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

This patent is subject to a terminal disclaimer.

(21) Appl. No.: **18/660,553**

(22) Filed: **May 10, 2024**

(65) **Prior Publication Data**

US 2024/0288013 A1 Aug. 29, 2024

Related U.S. Application Data

(63) Continuation of application No. 17/542,200, filed on Dec. 3, 2021, now Pat. No. 12,012,980, which is a (Continued)

(51) **Int. Cl.**
F15C 1/18 (2006.01)
B01F 25/4314 (2022.01)
(Continued)

(52) **U.S. Cl.**
CPC .... ***F15C 1/18*** (2013.01); ***B01F 25/4314*** (2022.01); ***B01F 25/432*** (2022.01); (Continued)

(58) **Field of Classification Search**
CPC .... F15C 1/18; B01F 5/0614; B01F 5/064; B01F 5/0651; B01F 5/0659; B01F 5/12; (Continued)

(56) **References Cited**

U.S. PATENT DOCUMENTS

1,496,345 A 6/1924 Lichtenhaeler
1,500,103 A 7/1924 Burdon et al.
(Continued)

FOREIGN PATENT DOCUMENTS

CN 111536051 8/2022
DE 10054669 2/2002
(Continued)

OTHER PUBLICATIONS

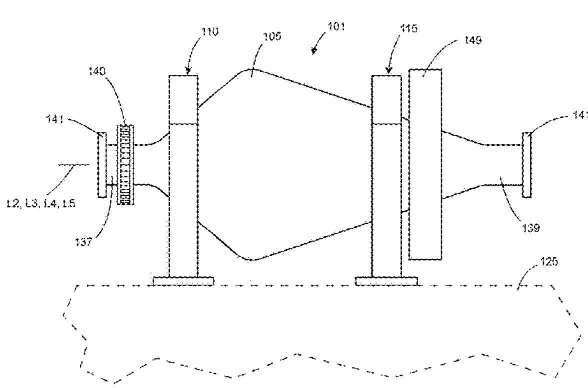
International Search Authority, Notification of Transmittal of the International Search Report and Written Opinion of the International Search Authority, PCT/US2019/0151468, 14 pages.
(Continued)

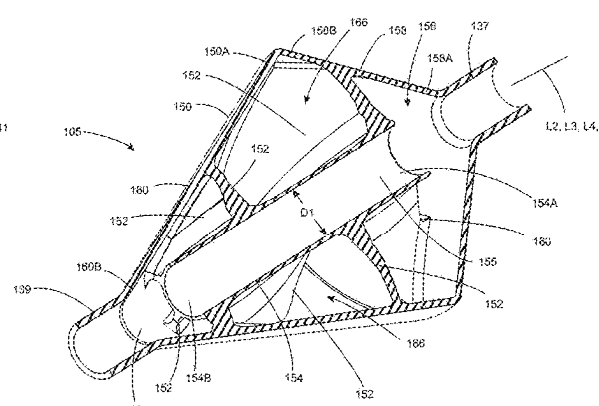
Primary Examiner — Charles Cooley
(74) *Attorney, Agent, or Firm* — David O. Simmons; IVC Patent Agency

(57) **ABSTRACT**

Disclosed cyclonic flow-inducing pumps overcome drawbacks associated with known adverse flow conditions that arise from flow of certain types of materials through a material flow conduit. Such cyclonic flow-inducing pumps provide for flow of flowable material within a flow passage of a material flow conduit (e.g., a portion of a pipeline, tubing or the like) to have a cyclonic flow (i.e., vortex or swirling) profile. Advantageously, the cyclonic flow profile centralizes flow toward the central portion of the flow passage, thereby reducing magnitude of laminar flow. Such cyclonic flow profile provides a variety of other advantages as compared to a parabolic flow profile such as, for example, increased flow rate, reduce inner pipeline wear, more uniform inner pipe wear, reduction in energy consumption, reduced or eliminated adverse considerations such as slugging.

18 Claims, 10 Drawing Sheets





Related U.S. Application Data

continuation-in-part of application No. 17/462,896, filed on Aug. 31, 2021, now Pat. No. 11,221,028, which is a continuation of application No. 16/991, 270, filed on Aug. 12, 2020, now Pat. No. 11,391,309, which is a continuation of application No. 16/846, 474, filed on Apr. 13, 2020, now Pat. No. 10,895,274, which is a continuation of application No. 16/567, 379, filed on Sep. 11, 2019, now Pat. No. 10,683,881, which is a continuation of application No. 16/445, 127, filed on Jun. 18, 2019, now Pat. No. 10,458,446.

(60) Provisional application No. 63/125,556, filed on Dec. 15, 2020, provisional application No. 62/917,233, filed on Nov. 29, 2018.

(51) **Int. Cl.**

B01F 25/432	(2022.01)
B01F 25/433	(2022.01)
B01F 25/435	(2022.01)
B01F 25/60	(2022.01)
B01F 29/00	(2022.01)
B01F 29/25	(2022.01)

(52) **U.S. Cl.**

CPC ***B01F 25/4334*** (2022.01); ***B01F 25/435*** (2022.01); ***B01F 25/60*** (2022.01); ***B01F 29/252*** (2022.01); ***B01F 29/40113*** (2022.01); *B01F 29/40221* (2022.01)

(58) **Field of Classification Search**

CPC .. B01F 7/00541; B01F 9/0007; B01F 9/0009; B01F 9/001; B01F 9/0041; B01F 2009/0092; B01F 25/4314; B01F 25/432; B01F 25/4334; B01F 25/435; B01F 25/60; B01F 29/252; B01F 29/40113; B01F 29/40221; F04D 3/00

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

1,513,624	A	10/1924	Parker	
1,777,141	A	9/1930	Howden	
1,959,907	A	5/1934	Ebert	
1,974,110	A	9/1934	Higley	
2,274,599	A	2/1942	Freeman	
2,300,130	A *	10/1942	McCurdy	F01N 1/12 181/274
2,784,948	A	3/1957	Pahl et al.	
2,816,518	A	12/1957	Daggett	
2,831,754	A *	4/1958	Manka	B01D 11/0473 423/658.5
4,204,775	A	5/1980	Speer	
4,339,918	A	7/1982	Michikawa	
5,743,637	A	4/1998	Ogier	
5,909,959	A	6/1999	Gerich	
5,992,465	A	11/1999	Jansen	
8,033,714	B2	10/2011	Nishioka et al.	
8,864,367	B2 *	10/2014	Hanada	B01F 5/0647 366/339
10,092,886	B2	10/2018	Kashihara et al.	
10,093,953	B2 *	10/2018	Gordon	C12P 7/04
10,201,786	B2 *	2/2019	Okada	B01F 5/0644
10,272,402	B2 *	4/2019	Tun	B01F 3/18
10,302,104	B2 *	5/2019	Schmidt	F16L 55/02772
10,458,446	B1 *	10/2019	Schmidt	B01F 5/0651
10,683,881	B1 *	6/2020	Schmidt	B01F 5/0614
10,844,887	B1 *	11/2020	Schmidt	E03F 3/02
10,890,200	B2 *	1/2021	Schmidt	B01F 5/0659
10,895,274	B2 *	1/2021	Schmidt	B01F 5/0614
11,002,301	B1 *	5/2021	Schmidt	F15D 1/065
11,085,470	B2 *	8/2021	Leutwyler	F04B 53/16
11,221,028	B1 *	1/2022	Schmidt	B01F 29/40113
11,247,758	B1 *	2/2022	Schmidt	B63H 11/103
11,319,974	B2 *	5/2022	Schmidt	B01F 25/432
11,378,110	B1 *	7/2022	Schmidt	F15D 1/02
11,391,309	B2 *	7/2022	Schmidt	B01F 25/435
11,460,056	B2 *	10/2022	Schmidt	F16L 57/06
11,713,764	B1 *	8/2023	Schmidt	F04D 13/08 415/173.1
11,739,774	B1 *	8/2023	Schmidt	F15D 1/0015 138/37
12,006,957	B2 *	6/2024	Schmidt	B01F 25/435
12,012,980	B2 *	6/2024	Schmidt	B01F 25/4334
2010/0307830	A1	12/2010	Curlett	
2012/0285173	A1	11/2012	Poyyapakkam et al.	
2016/0270893	A1	9/2016	Tapocik	
2017/0306994	A1 *	10/2017	Schmidt	F16L 55/02772
2019/0242413	A1 *	8/2019	Schmidt	F16L 43/00
2020/0173467	A1 *	6/2020	Schmidt	B01F 5/0614
2020/0173468	A1 *	6/2020	Schmidt	B01F 5/0614
2020/0263712	A1 *	8/2020	Schmidt	B01F 5/0659
2020/0370572	A1 *	11/2020	Schmidt	B01F 5/064
2020/0370573	A1 *	11/2020	Schmidt	F16L 55/02772
2021/0071692	A1 *	3/2021	Schmidt	F16L 55/02772
2021/0396252	A1 *	12/2021	Schmidt	B01F 25/60
2022/0090613	A1 *	3/2022	Schmidt	B01F 25/432
2022/0299049	A1 *	9/2022	Schmidt	B01F 25/432
2024/0288012	A1 *	8/2024	Schmidt	B01F 25/435
2024/0288013	A1 *	8/2024	Schmidt	B01F 25/60
2024/0410369	A1 *	12/2024	Schmidt	F04D 29/548

FOREIGN PATENT DOCUMENTS

EP	1134476	A1	9/2001	
FR	2614554	A1 *	11/1988	B01F 5/0057
GB	747576	A *	4/1956	B28C 5/1806
GB	971918		10/1964	
GB	971918	A *	10/1964	B01F 9/025
GB	2312276	A	10/1997	
KR	100803918	B	2/2008	
RU	2670283	C1	10/2018	

OTHER PUBLICATIONS

International Search Authority, Notification of Transmittal of the International Search Report and Written Opinion of the International Search Authority, PCT/US2021/061866, 8 pages.

European Patent Office, Extended European Search Report, 21907462.2-1004, 10 pages.

* cited by examiner

FIG. 1
-PRIOR ART-

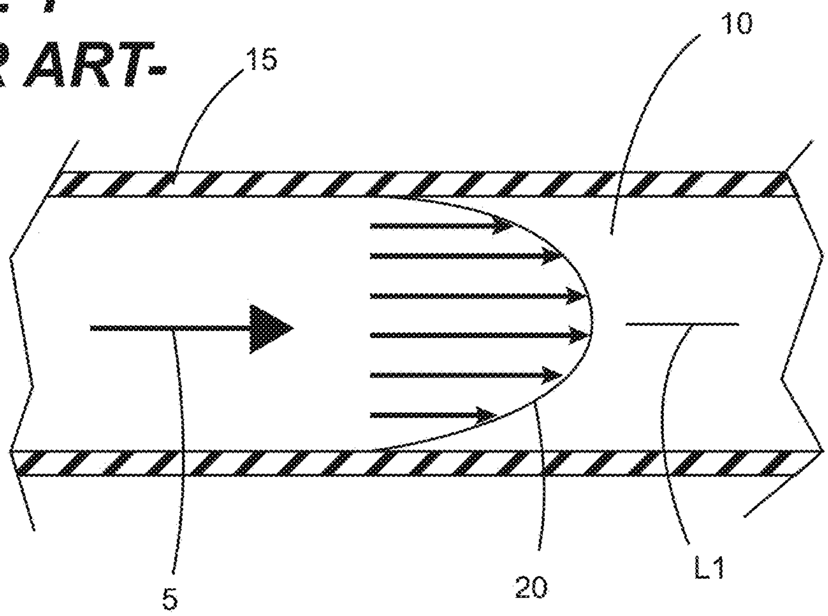


FIG. 2

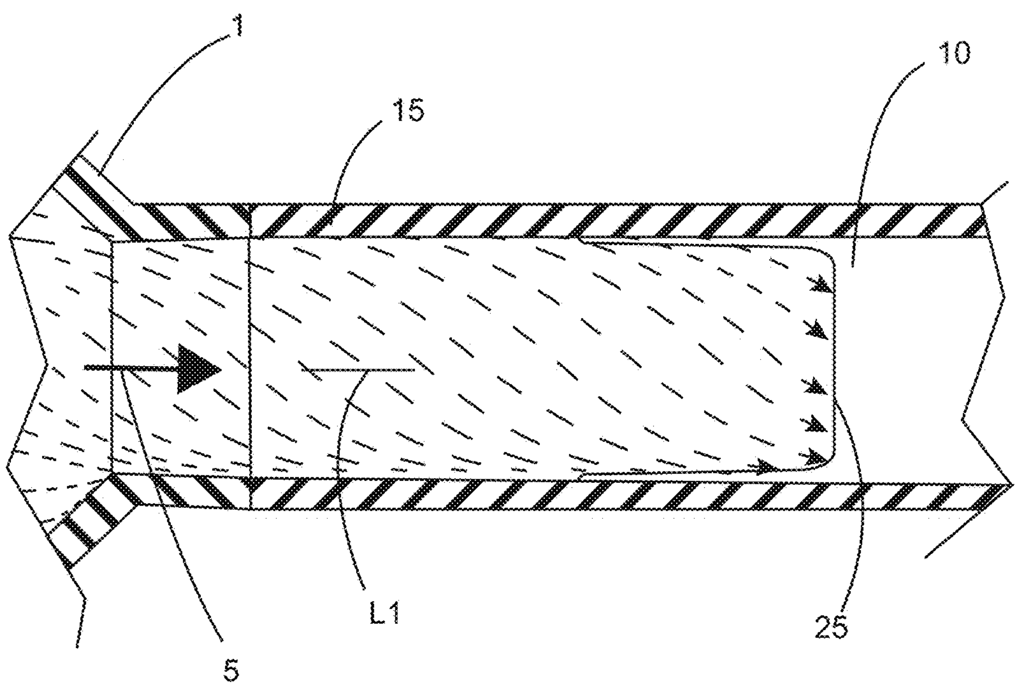


FIG. 3

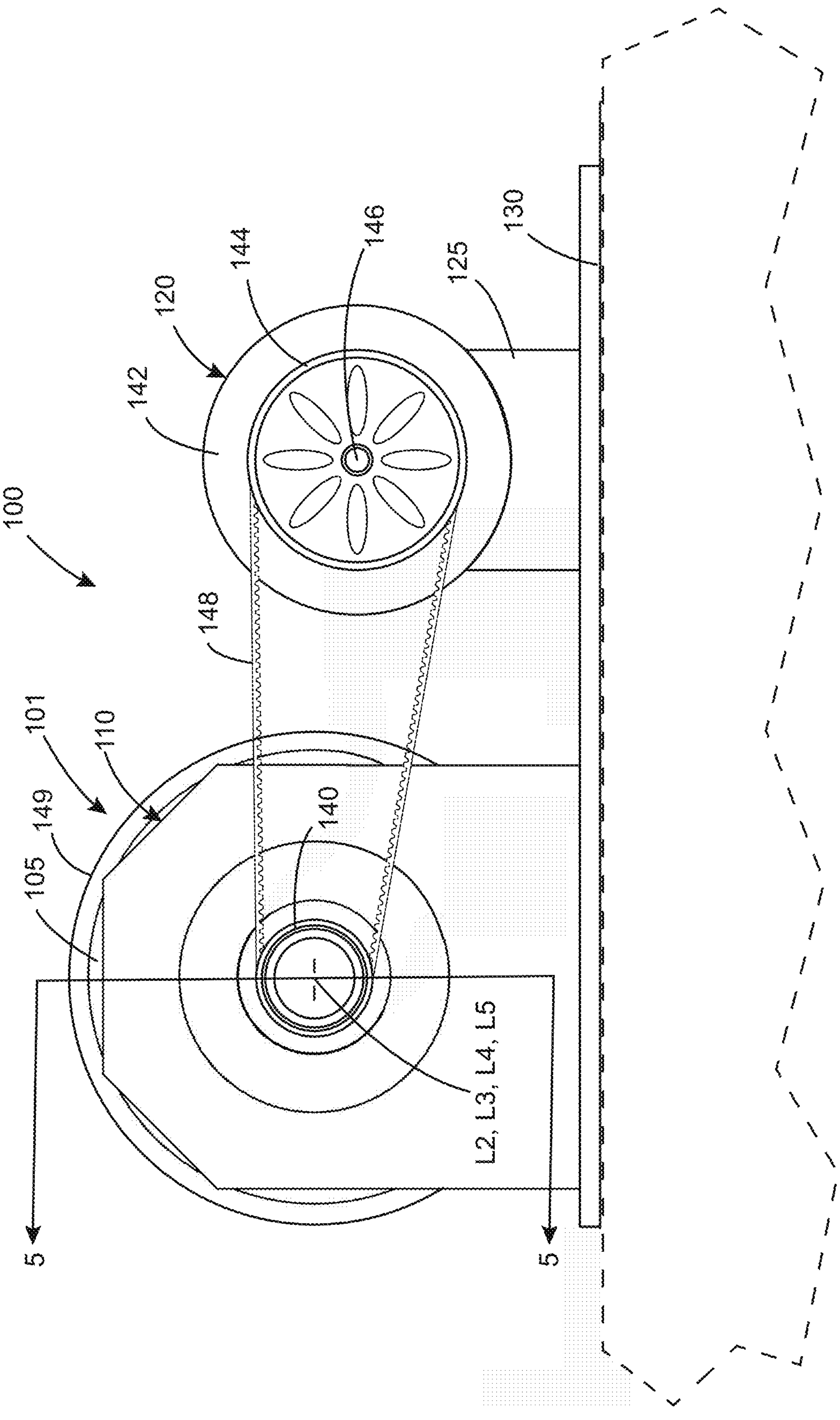


FIG. 4

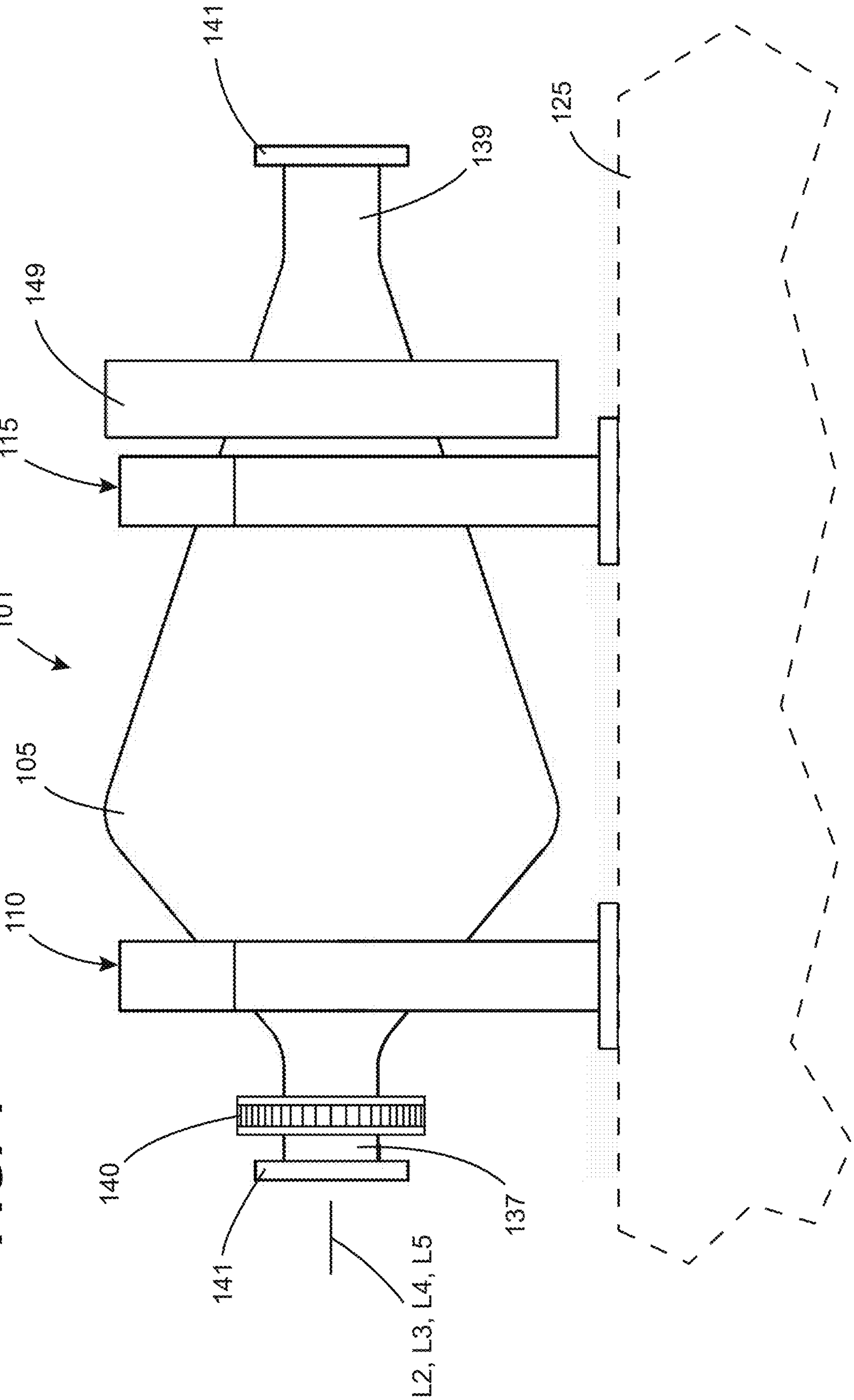
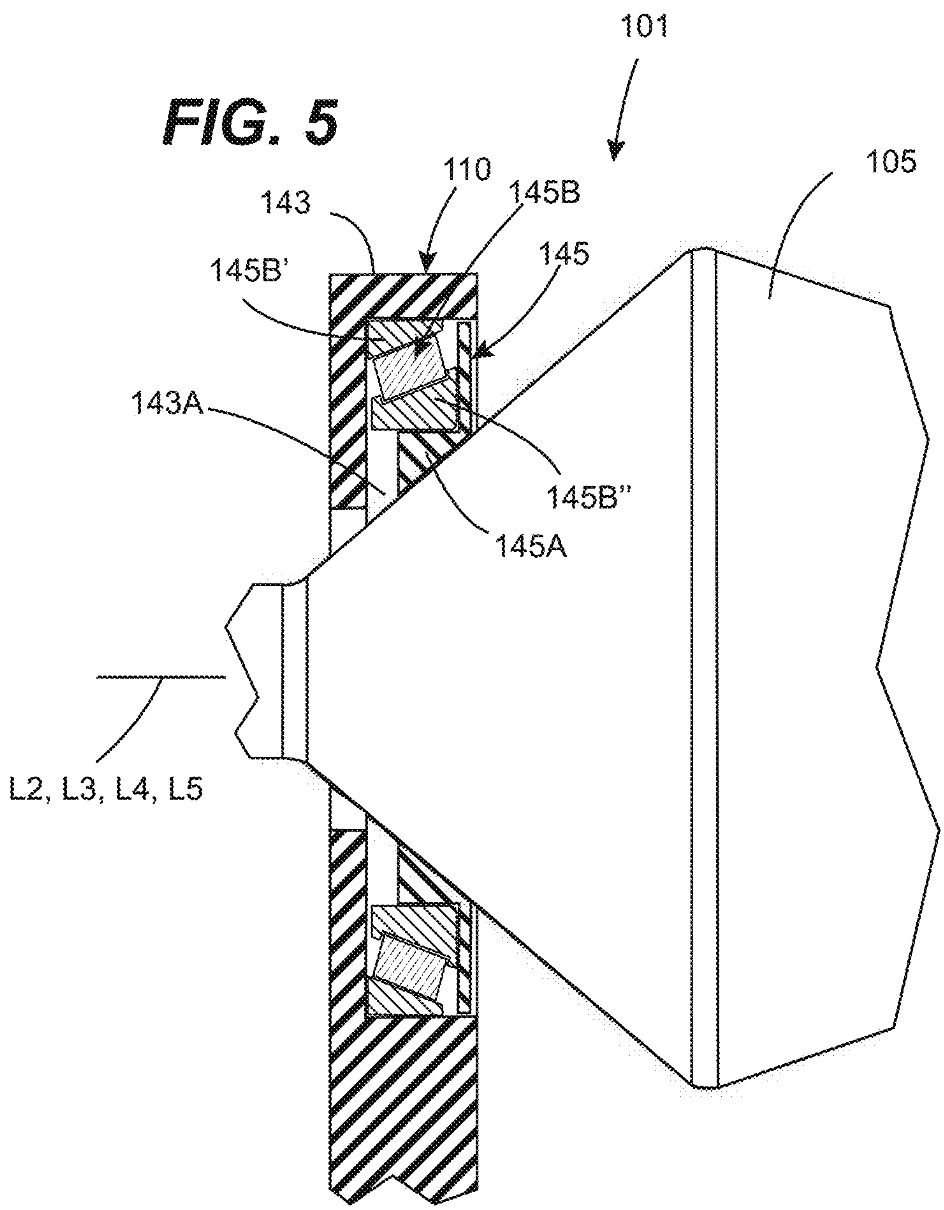


FIG. 5



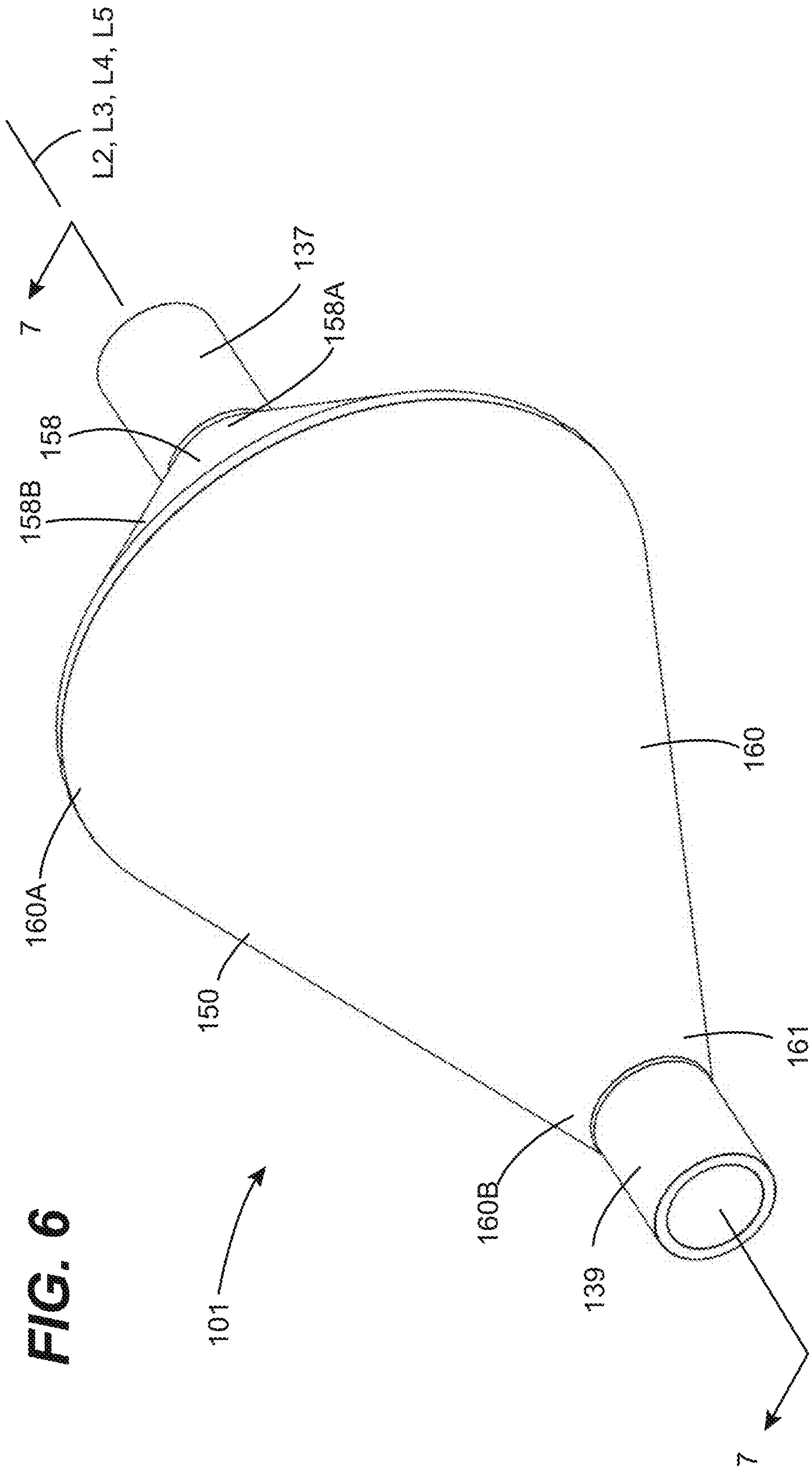


FIG. 6

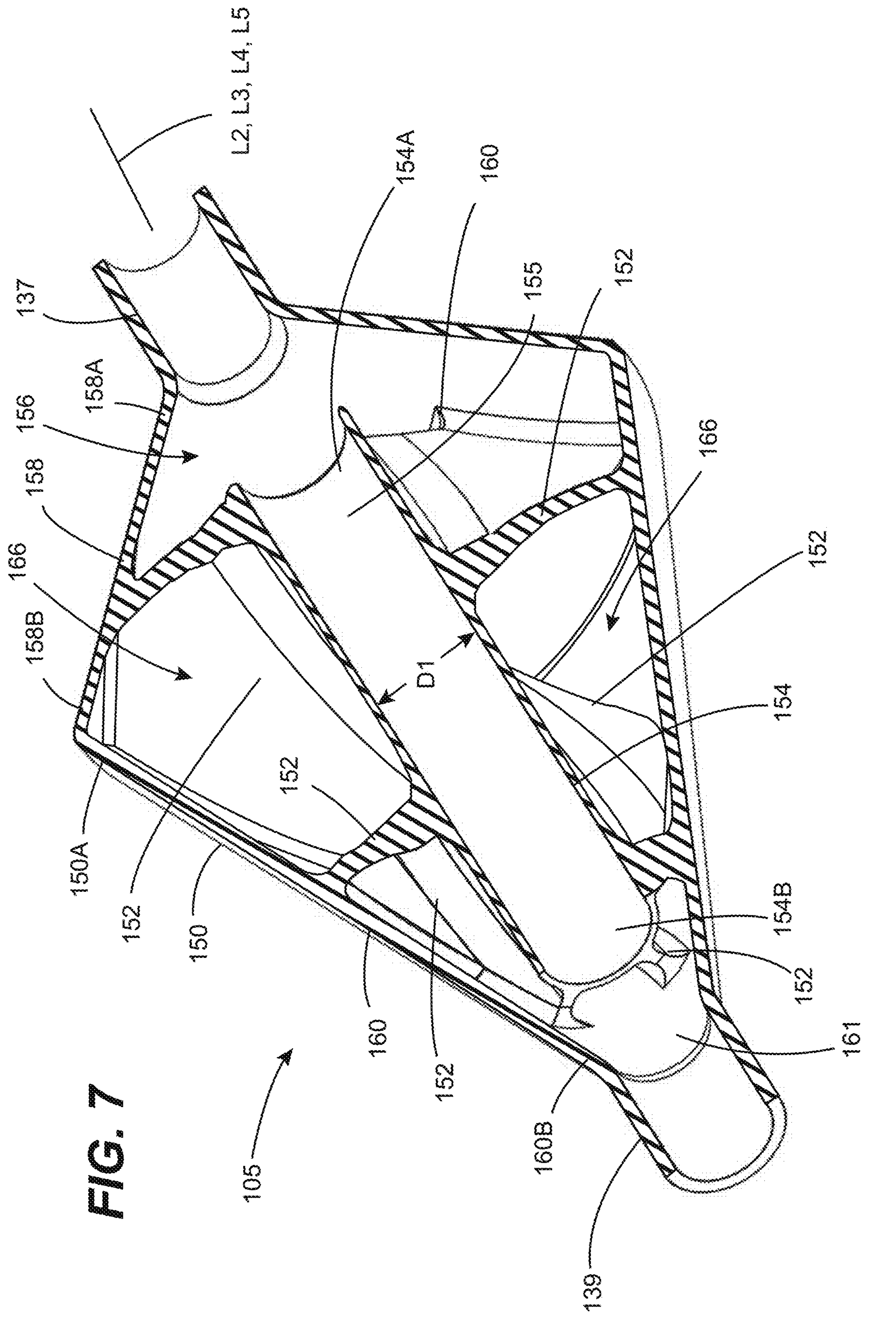

L2, L3, L4, L5
137
158A
156
158
166
158B
150A
152
150
105
154A
160
155
152
166
152
D1
154
152
154B
152
161
160
152
160B
139

FIG. 7

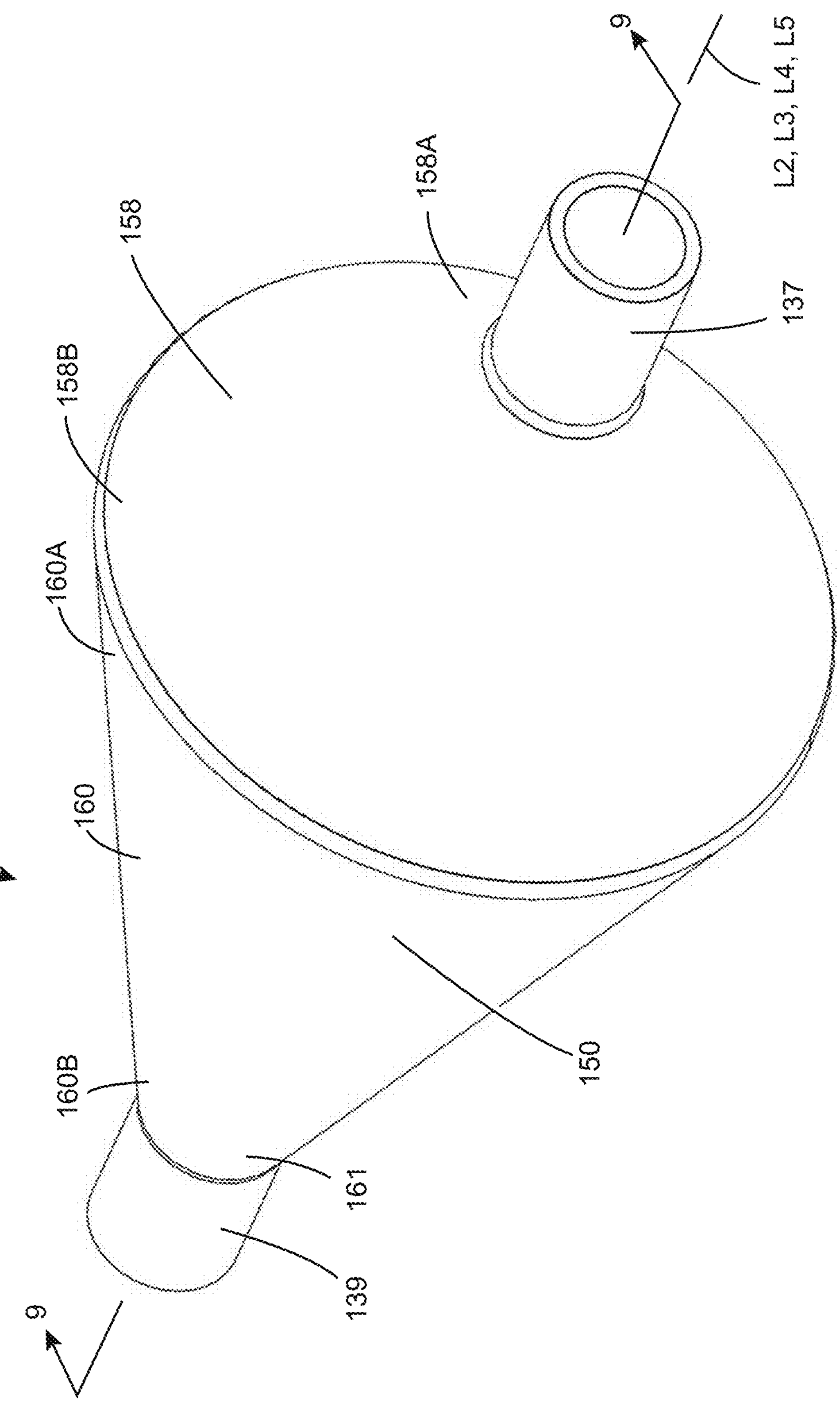


FIG. 8

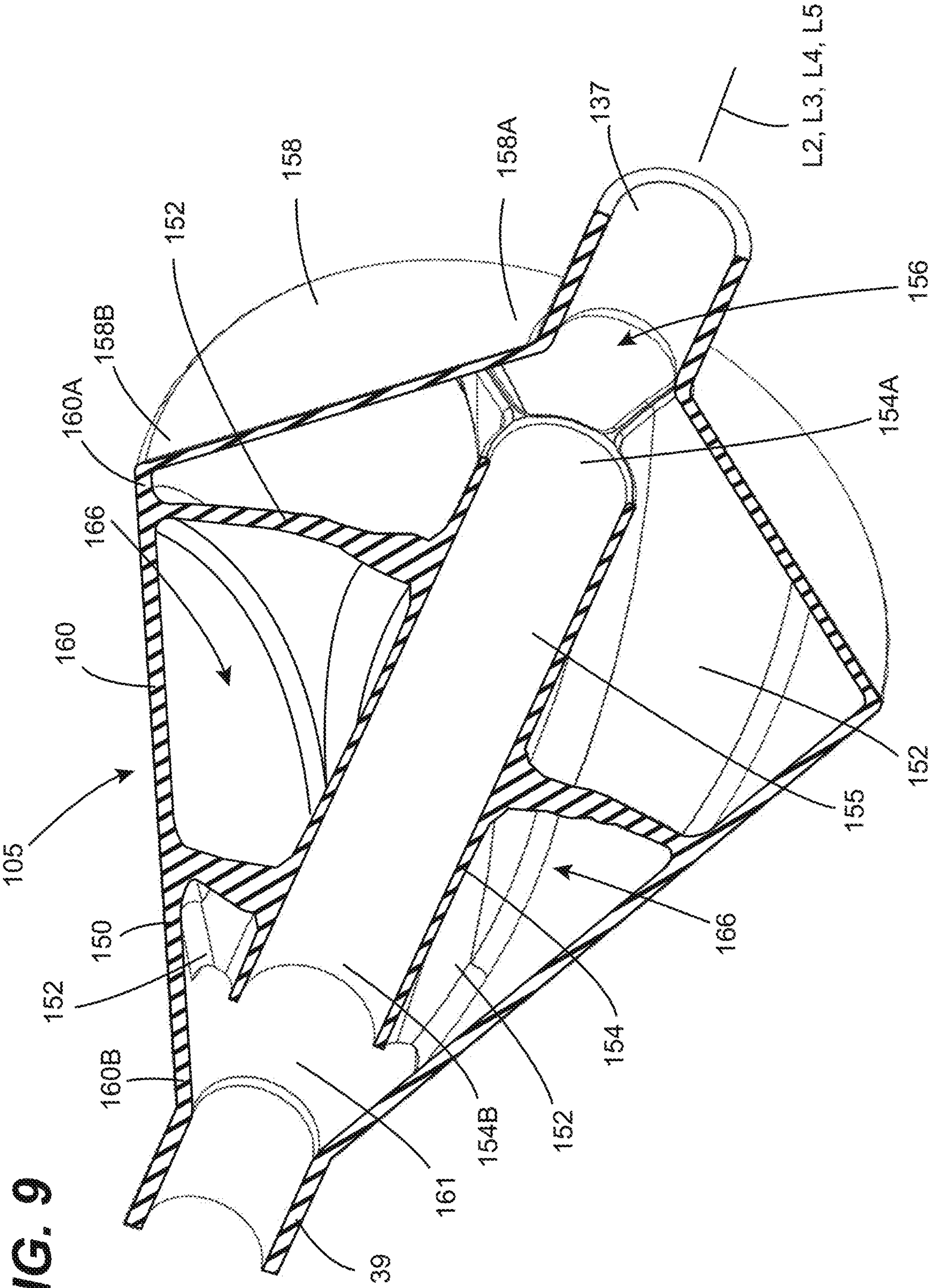
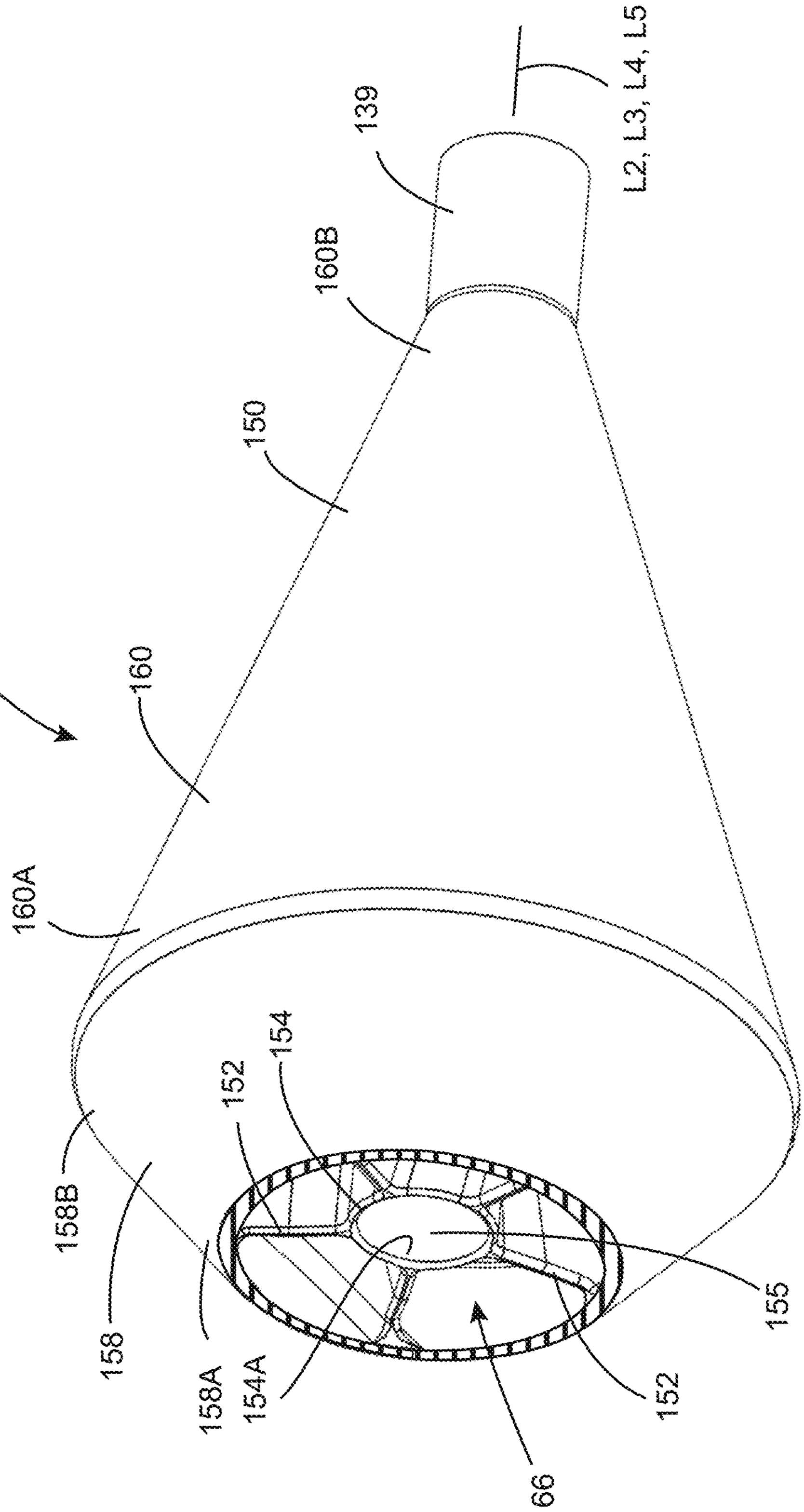

L2, L3, L4, L5
137
158A
158
152
156
158B
160A
154A
166
160
105
150
152
155
152
166
154
160B
152
154B
152
161
139

FIG. 9

FIG. 10


L2, L3, L4, L5
139
160B
150
105
160
160A
154
152
158B
158
158A
154A
155
152
166

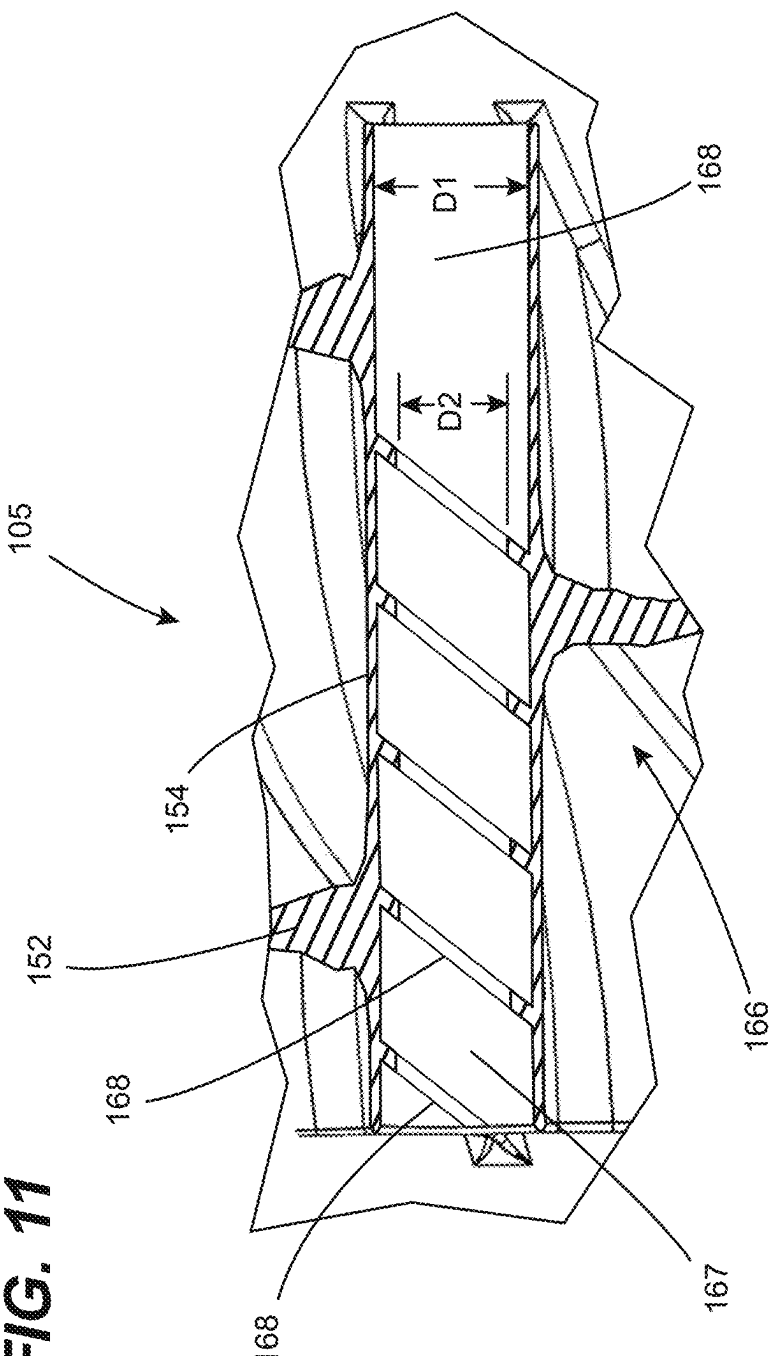


FIG. 11

CYCLONIC FLOW-INDUCING PUMP

CROSS-REFERENCE TO RELATED APPLICATIONS

This patent application is a continuation of U.S. Non-Provisional patent application Ser. No. 17/542,200, filed 3 Dec. 2021, entitled "CYCLONIC FLOW-INDUCING PUMP", now U.S. Pat. No. 12,012,980, which claims priority as a continuation of U.S. Non-Provisional patent application Ser. No. 17/462,896, filed 31 Aug. 2021, entitled "ROTARY IN-LINE PUMP", now U.S. Pat. No. 11,221,028, which claims priority as a continuation-in-part from U.S. Non-Provisional patent application Ser. No. 16/991,270, filed 12 Aug. 2020, entitled "MATERIAL FLOW AMPLIFIER", now U.S. Pat. No. 11,391,309 and from U.S. Provisional Patent Application No. 63/125,556, filed Dec. 15, 2020, entitled "ROTARY IN-LINE PUMP," each having a common applicant herewith and being incorporated herein in their entirety by reference. Non-Provisional patent application Ser. No. 16/991,270 claims priority from U.S. Non-Provisional patent application Ser. No. 16/846,474, filed 13 Apr. 2020, now U.S. Pat. No. 10,895,274, entitled "MATERIAL FLOW AMPLIFIER", having a common applicant herewith and being incorporated herein in its entirety by reference. Non-Provisional patent application Ser. No. 16/846,474 claims priority as continuation patent application from U.S. Non-Provisional patent application Ser. No. 16/567,379, filed 11 Sep. 2019, now U.S. Pat. No. 10,683,881, entitled "MATERIAL FLOW AMPLIFIER", having a common applicant herewith and being incorporated herein in its entirety by reference. Non-Provisional patent application Ser. No. 16/567,379 claims priority as continuation patent application from U.S. Non-Provisional patent application Ser. No. 16/445,127, filed 18 Jun. 2019, now U.S. Pat. No. 10,458,446, entitled "MATERIAL FLOW AMPLIFIER", having a common applicant herewith and being incorporated herein in its entirety by reference. U.S. Non-Provisional patent application Ser. No. 16/445,127 claims priority from co-pending United States Provisional Patent Application having Ser. No. 62/917,233, filed 29 Nov. 2018, entitled "MULTI-CHAMBERED VORTEX PIPELINE AMPLIFIER (FULLY PIGGABLE)", having a common applicant herewith and being incorporated herein in its entirety by reference.

FIELD OF THE DISCLOSURE

The disclosures made herein relate generally to pumps for flowable materials and, more particularly, to rotary in-line pumps for flowable materials and preferably for liquid materials.

BACKGROUND

The need to generate pressurized flow by imparting a pumping action on a flowable material or other suitably viscous form of material through a material flow conduit (i.e., a pumpable material) is well known. Examples of such material flow conduit include, but are not limited to, pipes, pipelines, conduits, tubular flow members, and the like. The pumping action is provided by a pump to increase a pressure of the flowable material at an outlet of the pump and thereby increases flow velocity of the flowable material downstream of the pump.

As shown in FIG. **1**, conventional flow of flowable material **5** within a flow passage **10** of material flow conduit **15** has a flow profile characterized by laminar flow effect (i.e., laminar flow **20**). The parabolic flow profile is a result of the laminar boundary layer along the surface of the material flow conduit **15** defining the flow passage **10**. Flowable material at the surface of the flow passage **10** exhibits considerable friction and zero flow velocity, thereby reducing velocity of the flowable material even at a considerable distance from the surface of the flow passage **10**. In association with this reduced velocity, the laminar flow effect (e.g., friction at the surface of the material flow conduit) is known to increase head loss and heating of the flowable material.

There are various well-known flow considerations that arise when a flowable material and, particularly abrasive flowable material, flows through a material flow conduit such as a pipeline. One such consideration is erosion (i.e., wearing) of the material flow conduit. Transport and pumping flowable material comprising abrasive contents, such as coal and sand slurries, wet sand, gravel and the like can cause especially high costs associated with component wear due to interaction between the flowable material and the surface defining the passage through which such material flows. Additionally, uneven erosion in piping systems, especially elbow fittings, is well known to lead to fitting failure or early fitting replacement, either of which is costly in material, manpower and downtime.

When flowable materials pass through an elbow fitting, the change in direction creates turbulent conditions, flow separation and vortex shedding along the pipe wall at the inside of the bend of the elbow fitting. This change in direction may also create standing eddies causing backflow conditions at points along the elbow fitting pipe walls. These conditions generally cause the elbow fitting pipe wall along the outside of the bend to erode substantially faster than the pipe wall along the inside of the bend because the flowable material impinges directly against the wall along the outside of the bend as it enters the fitting and changes direction. Additionally, due to centrifugal force, heavier solids and particulates are generally thrown to the outside wall as the flowable material changes direction and tend to continually scour the outer wall.

A similar uneven erosion effect is often experienced in straight pipe runs. For example, the concentration of particulates of a flowable material will increase in the lower region of the fluid in long straight runs (i.e., particulate dropping out of suspension), making the bottom portion of the fluid stream more abrasive or prone to material deposition and/or aggregation than the upper portion. Such material deposition and/or aggregation can alter fluid flow conditions (e.g., velocity, temperature, pressure and the like) and can alter the material composition of the flowable material (e.g., less downstream concentration of particular material than required or intended). Additionally, in large diameter piping systems, the weight of the flowable material is borne by the lower pipe wall portion thereby causing higher erosion rates.

Another well-known flow consideration that arises is head loss due to turbulence and flow separation in an elbow fitting. Higher pumping pressures can be utilized for mitigating this head loss resulting from such head losses. However, higher pumping pressures are generally implemented at the expense of higher energy consumption and associated cost. Additionally, implementation of higher pumping pressures often creates vibration and heating problems in the piping system.

Long radius elbow fittings and pipe sections can reduce these adverse flow considerations. However, long radius fittings require a great deal of space relative to standard (i.e., short) radius fittings. Additionally, long radius fittings still suffer accelerated erosion rates along the pipe wall along the outside of the bend because centrifugal force still causes heavier, more abrasive flowable materials to be thrown to the outer wall, and they are continually scoured by on-going flow of such flowable material.

Therefore, a pump adapted to produce material flow characteristics that overcome drawbacks associated with known flow considerations that arise from flow of flowable materials flowing through a material flow conduit would be beneficial, desirable and useful.

SUMMARY OF THE DISCLOSURE

Embodiments of the disclosures made herein are directed to a pump adapted to produce material flow characteristics that overcome drawbacks associated with known adverse flow conditions in pipe structures through a material flow conduit. In preferred embodiments, the pump is a rotary in-line pump that can be connected between two sections of material flow conduit. A pump in accordance with one or more embodiments of the disclosures made herein provides for flow of flowable material within a flow passage of a material flow conduit (e.g., a portion of a pipeline, tubing or the like) to have a cyclonic flow (i.e., whirlpool, vortex or rotational) profile. Advantageously, such a cyclonic flow profile centralizes flow toward the central portion of the flow passage, thereby reducing the magnitude of laminar flow. Such cyclonic flow profile is also known to provide a variety of other advantages as compared to a parabolic flow profile resulting from laminar flow—e.g., increased flow rate, reduce inner pipeline wear, more uniform inner pipe wear, reduction in energy consumption, reduced or eliminated slugging and the like.

In one or more embodiments of the disclosures made herein, a rotary in-line pump comprises a material pressurizer and a plurality of mounting units attached thereto. The material pressurizer including a plurality of helical flow passages each jointly defined by a respective portion of an exterior body, a respective portion of a centralizer tube and adjacent ones of a plurality of helical vanes that extended between the exterior body and the centralizer tube at least partially along a length of the exterior body. A longitudinal centerline axis of the exterior body and a longitudinal centerline axis of the centralizer tube extend colinearly with a longitudinal reference axis. Each of the helical flow passages includes a divergent portion having increasing cross-sectional area along a first portion of a length of the exterior body and a convergent portion having decreasing cross-sectional area along a second portion of a length of the exterior body. The convergent portion is in fluid communication with and extends from the divergent portion such that each of the helical flow passages is contiguous along a length thereof. The plurality of mounting units each have a support body and a bearing assembly. An upstream one of the mounting units has the bearing assembly thereof engaged with an upstream portion of the exterior body. A downstream one of the mounting units has the bearing assembly thereof engaged with a downstream portion of the exterior body. A centerline longitudinal axis of the bearing assembly of the upstream one of the mounting units and a centerline longitudinal axis of the bearing assembly of the downstream one of the mounting units each extend colinearly with the longitudinal reference axis thereby enabling the material pressurizer to rotate in a radially constrained manner about the longitudinal reference axis.

In one or more embodiments of the disclosures made herein, a rotary in-line pump system comprises a material pressurizer, a plurality of mounting units and a drive apparatus. The material pressurizer includes an exterior body, a plurality of helical vanes within an interior space of the exterior body and a centralizer tube within the interior space of the exterior body. The exterior body includes a conically divergent section and a conically convergent section. The conically divergent section and the conically convergent section each have an upstream end portion and a downstream end portion. The downstream end portion of the conically divergent section is attached to the upstream end portion of the conically convergent section. A longitudinal centerline axis of the conically divergent section, a longitudinal centerline axis of the conically convergent section and a longitudinal centerline axis of the centralizer tube extend colinearly with a longitudinal reference axis. Each of the helical vanes extends between an interior surface of the exterior body and an exterior surface of the centralizer tube at least partially along a length of the exterior body to define a plurality of helical flow passages extending between the exterior body, the centralizer tube and adjacent ones of the helical vanes. The plurality of mounting units each have a support body and a bearing assembly. An upstream one of the mounting units has the bearing assembly thereof engaged with the conically divergent section of the exterior body. A downstream one of the mounting units has the bearing assembly thereof engaged with the conically convergent section of the exterior body. A centerline longitudinal axis of the bearing assembly of the upstream one of the mounting units and a centerline longitudinal axis of the bearing assembly of the downstream one of the mounting units each extend colinearly with the longitudinal reference axis thereby enabling the material pressurizer to rotate about the longitudinal reference axis. The drive apparatus is engaged with the material pressurizer and exerts rotational force on the material pressurize for causing the material pressurizer to rotate about the longitudinal reference axis.

In one or more embodiments, the centralizer tube has a uniform outside diameter.

In one or more embodiments, the centralizer tube has a cylindrical cross-sectional profile.

In one or more embodiments, the exterior body has a first tapered section defining a profile of the divergent portion of each of the helical flow passages and a second tapered portion defining a profile of the convergent portion of each of the helical flow passages.

In one or more embodiments, the exterior body includes a conically divergent section and a conically convergent section, the conically divergent section and the conically convergent section each have an upstream end portion and a downstream end portion, the downstream end portion of the conically divergent section is attached to the upstream end portion of the conically convergent section, and a longitudinal centerline axis of the conically divergent section and a longitudinal centerline axis of the conically convergent section each extend colinearly with the longitudinal reference axis.

In one or more embodiments, each of the helical flow passages extends along an entire length of the centralizer tube.

In one or more embodiments, each of the helical flow passages and a central passage of the centralizer tube terminate at a flow mixer section of the material pressurizer.

In one or more embodiments, an inside diameter of the centralizer tube is uniform over an entire length of thereof.

In one or more embodiments, an inside diameter of the centralizer tube is non-uniform over at least a portion of a length of thereof.

In one or more embodiments, at least one helical projection is on an interior surface of the central passage of the centralizer tube.

In one or more embodiments, the at least one helical projection extends at least partially along an entire length of the centralizer tube.

In one or more embodiments, a pitch of the at least one helical projection is the same as a pitch of each of the helical flow passages.

In one or more embodiments, a maximum inside diameter of the exterior body is at least 4 times the inside diameter of the centralizer tube and a minimum inside diameter of the exterior body is approximately the same as the inside diameter of the centralizer tube.

In one or more embodiments, an inlet of each of the helical flow passages and an inlet of the centralizer tube all lie in a common plane.

In one or more embodiments, the exterior body includes a material inlet body at an upstream end portion thereof and a material outlet body at a downstream end portion thereof.

In one or more embodiments, a longitudinal centerline axis of the material inlet body and a longitudinal centerline axis of the material outlet body each extend colinearly with the longitudinal reference axis.

In one or more embodiments, the material inlet body, the material outlet body and the centralizer tube each have a common inside diameter that is uniform over an entire length of thereof.

In one or more embodiments, the material outlet body and the centralizer tube each have a common inside diameter that is uniform over an entire length of thereof.

In one or more embodiments, at least about 80% of a length of the centralizer tube resides within an interior space of the exterior body defined by a conically convergent section thereof.

These and other objects, embodiments, advantages and/or distinctions of the present invention will become readily apparent upon further review of the following specification, associated drawings and appended claims.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a diagrammatic view showing laminar flow effect within a material flow conduit.

FIG. **2** is a diagrammatic view showing conversion from a laminar flow effect to cyclonic flow effect by a rotary in-line pump configured in accordance with one or more embodiments of the disclosures made herein.

FIG. **3** is a diagrammatic end view showing a rotary in-line pump system configured in accordance with one or more embodiments of the disclosures made herein.

FIG. **4** is a diagrammatic side view showing a rotary in-line pump of the rotary in-line pump system of FIG. **3**.

FIG. **5** is a fragmentary cross-sectional view taken along the line **5-5** in FIG. **3**.

FIG. **6** is a first perspective view showing a material pressurizer configured in accordance with one or more embodiments of the disclosures made herein.

FIG. **7** is a cross-sectional view taken along the line **7-7** in FIG. **6**.

FIG. **8** is a second perspective view of the material pressurizer shown in FIG. **6**.

FIG. **9** is a cross-sectional view taken along the line **9-9** in FIG. **8**.

FIG. **10** is a cross-sectional view showing an upstream end portion of helical flow passages of the material pressurizer shown in FIG. **6**.

FIG. **11** is a cross-sectional fragmentary view showing a material pressurizer configured in accordance with one or more embodiments of the disclosures made herein, wherein a centralizer tube of the material pressurizer has a plurality of helical projections provided on an interior surface defining a central passage thereof.

DETAILED DESCRIPTION

Embodiments of the disclosures made herein are directed to rotary in-line pumps that provide for increased volumetric flow rates for flowable material (e.g., fluids, slurries, and the like) and for associated reductions in wear to material flow conduits through which flow of such flowable materials is provided. Rotary in-line pumps in accordance with embodiment of the disclosures made herein induce a cyclonic flow profile (i.e., rotational, vortex or swirling movement) that advantageously overcomes drawbacks associated with known adverse flow conditions (e.g., internal pipe wall erosion, head losses, material heating) that can arise from flow of various types of flowable materials flowing through a material flow conduit in a conventional manner (e.g., under laminar flow effect). Rotary in-line pumps in accordance with embodiment of the disclosures made herein can be mounted and operated in any angular orientation (i.e., omni-directionally mountable).

As discussed above in reference to FIG. **1**, conventional of flowable material **5** within a flow passage **10** of a material flow conduit **15** has a flow profile characterized by laminar flow effect (i.e., laminar flow **20**). However, advantageously, a rotary in-line pump **1** in accordance with one or more embodiments of the disclosures made herein is configured in a manner that causes conventional flow to be transformed from a flow profile characterized by laminar flow effect to a flow profile being characterized by cyclonic flow effect (i.e., cyclonic flow **25**). Cyclonic flow effect is the result of cyclonic movement of the flowable material **5** (i.e., sometime also referred to as cyclonic, whirlpool or vortex flow effect) about the longitudinal axis L**1** of the material flow conduit **15** as generated by the rotary in-line pump **1**. As a person or ordinary skill in the art will understand (e.g., as depicted in FIGS. **1** and **2**), cyclonic flow provides greater average flow velocity and volumetric flow than laminar flow for a given material flow conduit. Additionally, cyclonic flow mitigates adverse interaction between the surface of the material flow conduit and the flowable material. These advantageous aspects of cyclonic flow arise from the cyclonic flow profile accelerating and centralizing flow of the flowable material toward the central portion of the flow passage **10**, thereby mitigating associated adverse flow conditions and amplifying flow magnitude.

As will also become apparent from the disclosures made herein, a rotary in-line pump in accordance with embodiments of the disclosures made herein advantageously drives flowable material flow toward a focal point along a centerline axis of the rotary in-line pump (and thus the downstream flow conduit). Without this focal point functionality, material flow leaving the rotary in-line pump would be that of a centrifuge—i.e., material being undesirably accelerated and driven toward the interior surface of the material flow conduit. In contrast, by driving the flowable material toward the centerline axis of the material flow conduit via helical flow passages, the amount of flowable material at the interior surface of the material flow conduit is greatly reduced as compared to laminar flow or centrifuge-induced flow. Additionally, by driving flowable material flow toward the focal point of the rotary in-line pump, a portion of the flowable material (i.e., generally non-rotating flowable material) can become trapped between the inside surface of the material flow conduit (e.g., pipeline) and the exterior boundary of the rotationally flowing flowable material, thereby becoming an interface material for the rotationally flowing flowable material that serves to lower the effective coefficient of friction exhibited at the exterior boundary of the rotationally flowing flowable material (i.e., flowing of flowable material upon like material as opposed to material of the material flow conduit). Accordingly, in view of the material flow being driven toward the centerline axis of the rotary in-line pump (i.e., toward the focal point of the rotary in-line pump), the cyclonic flow profile provided for by rotary in-line pumps in accordance with embodiments of the disclosures made herein is propagated along the material flow conduit (e.g., because a large amount of the side wall drag is eliminated) and wear is thus dramatically reduced.

To maintain the beneficial effects of cyclonic flow, one or more additional rotary in-line pumps can be provided downstream of an initial rotary in-line pump. The distance between rotary in-line pumps can be proportional to system attributes such as, for example, pipe size, volume of fluid desired flow rates, pipeline's layout, terrain (e.g., elevation grade) and the like. The objective of placement and configuration of the rotary in-line pump is to reduce side wall drag, thereby increasing flow and utilizing the full potential of the cross-sectional flow area of a material flow conduit.

In a conventional pipe structure, internal pipe wear occurs unevenly because of the concentration of wear particles scuffing the lowest area of the pipe. In a conventional piping system heavier particle fall out and drag along the bottom of the pipe structure. The cyclonic action provided for by rotary in-line pumps in accordance with the disclosures made herein keeps particles suspended. In all flow directional changes such as in elbow pipes, the same particles are thrown to the outside as if it were in a centrifuge. In contrast, cyclonic flow as provided for by rotary in-line pumps in accordance with one or more embodiments of the disclosures made herein acts to focus flowable material flow through more uniformly across the centerline and cross-section portion of the of the material flow conduit with less boundary layer contact. Thus, the use of one or more rotary in-line pumps in accordance with one or more embodiments of the disclosures made herein can mitigate uneven wear, erosion and the like within material flow conduit.

Referring now to FIGS. **3-5**, specific aspects of a rotary in-line pump system in accordance with one or more embodiments of the disclosures made herein (i.e., rotary in-line pump system **100**) are discussed. The rotary in-line pump system **100** includes a material pressurizer **105**, an upstream mounting unit **110**, a downstream mounting unit **115** and a drive apparatus **120**. The material pressurizer **105** and the mounting units **110**, **115** jointly define a rotary in-line pump **101**. The mounting units **110**, **115** and a mounting base **125** of the drive apparatus **120** can be fixedly attached, directly or indirectly, to a support structure **130** (prior art) such as, for example, a concrete foundation, mounting body of an article or the like. Such attachment has the objective of securing the rotary in-line pump **101** and the drive apparatus **120** in fixed positional relationship relative to each other and to a fixed geospatial position.

The mounting units **110**, **115** are each engaged with a respective portion of the material pressurizer **105** of the rotary in-line pump **101**. The upstream mounting unit **110** is engaged with an upstream portion of the material pressurizer **105** and the downstream mounting unit **115** is engaged with a downstream portion of the material pressurizer **105**. Engagement of the mounting units **110**, **115** with the respective portion of the material pressurizer **105** and the attachment of the mounting units **110**, **115** to the support structure **130** serves to fixedly constrain the material pressurizer **105** axially and radially with respect to a longitudinal reference axis L**2** and to enable rotation of the material pressurizer **105** about the longitudinal reference axis L**2**. In one or more embodiments, each of the mounting units **110**, **115** can include a support body **143** and a bearing assembly **145** that is coupled between the support body **143** thereof and a respective portion of the material pressurizer **105** to provide for enabling such axially and radially constraint of the material pressurizer **105** relative to the longitudinal reference axis L**2** and for enabling such rotation of the material pressurizer **105** about the longitudinal reference axis L**2**. The bearing assembly **145** (or rotation enabling structure of the mounting units **110**, **115**) preferably suitably limit (e.g., eliminate) endplay from the material pressurizer **105** relative to the mounting units **110**, **115**. For example, the bearing assembly **145** can include a bearing mount **145**A engaged with or unitary to the material pressurizer **105** and a bearing **145**B having an inner race portion **145**B' thereof mounted on the bearing mount **145**A. An exterior race portion **145**B" of the bearing **145**B is engaged with a bearing engaging portion **143**A of the support body **143** of the respective one of the mounting units **110**, **115**.

The material pressurizer **105** includes a pulley **140** (or other type of output energy receiving device) attached to a material inlet body **137** of the material pressurizer **105**. The pump **101** receives rotational energy from the drive apparatus **120** through the pulley **140**. The drive apparatus **120** includes a motor **142** (or other type of rotation force-generating device) and a pulley **144** (or other type of output energy device) attached to an output shaft **146** of the motor **140**. A drive member **148** of the drive apparatus **120** is engaged between the pulley **140** of the material pressurizer **105** and the pulley **144** of the drive apparatus **120** for enabling rotational energy to be conveyed from the motor **142** to the material pressurizer **105**.

It is disclosed that embodiments of the disclosures made herein are not limited to a particular type of configuration of drive apparatus. Drive apparatuses using a belt or the like for conveying rotational energy can be utilized as can be drive apparatuses using engaged rotating members such as gears, axles, and the like. Rotary in-line pump systems in accordance with embodiments of the disclosures made herein are not unnecessarily limited to a particular type or configuration of drive apparatus. In preferred embodiments, the rotational power can be delivered from the drive apparatus **120** to the pump **101** in a multiplied or reduced manner (e.g., preset or adjustable) whereby rotational speed of the material pressurizer **105** about the longitudinal reference axis L2 can be set at a desired level relative to an associated rotational speed of the output shaft **146** of the motor **142**. In some instances, it will be preferred to drive the material pressurizer **105** at a rotational speed greater than that of the output shaft **146** of the motor **142**. In other instances, it will be preferred to drive the material pressurizer **105** at a rotational speed less than that of the output shaft **146** of the motor **142**. In these regards, rotary in-line pump system in accordance with one or more embodiments of the disclosures made herein can be adapted to achieve preset or adjustable torque multiplication between the material pressurizer **105** and the output shaft **146** of the motor **142**.

As shown in FIGS. **3** and **4**, the rotary in-line pump **101** can include an inertial device **149** that is externally mounted on or unitary formed with the material pressurizer **105**. In one or more embodiments, the inertial device **149** can be a harmonic balancer, a flywheel, both a separate harmonic balancer and a separate flywheel or a combined device (i.e., a single unit) that provides the functionality of both the flywheel and the harmonic balancer. The harmonic balancer can be a portion of the inertial device **149** that is relatively close to the longitudinal reference axis L**2**. The objective of the harmonic balancer functionality is to eliminate all frequencies or at least critically adverse frequencies generated in the structure of the pump **101** up to at least a predetermined engineered gravitational force (i.e., gravitational force (32.2 ft/s^2)). In contrast, the objective of the flywheel functionality is to use inertial mass to help maintain rotational speed of the material pressurizer **105** (i.e., rotational momentum) by mitigating or eliminating rotational surges of the pump **101** that can arise from, for example, variable fluid densities, pipe volumes and the like.

Referring now to FIGS. **6-10**, specific aspects of the material pressurizer **105** (i.e., a material pressurizer in accordance with one or more embodiments of the disclosure made herein) are disclosed. The material pressurizer **105** includes an exterior body **150**, a plurality of helical vanes **152** and a centralizer tube **154**. In some embodiments, the exterior body **150** can be a tubular body—e.g., having generally uniform thickness walls. The helical vanes **152** and the centralizer tube **154** are located within an interior space **156** of the exterior body **150**. The exterior body **150** includes a conically divergent section **158** and a conically convergent section **160**. The conically divergent section **158** and the conically convergent section **160** each have an upstream end portion **158**A, **160**A and a downstream end portion **158**B, **160**B. The downstream end portion **158**B of the conically divergent section **158** is attached to the upstream end portion **160**A of the conically convergent section **160**. The centralizer tube **154** has an upstream end portion **154**A located within the conically divergent section **158** and a downstream end portion **154**B located within the conically convergent section **160**. A longitudinal centerline axis L**3** of the conically divergent section **158**, a longitudinal centerline axis LA of the conically convergent section **160** and a longitudinal centerline axis L**5** of the centralizer tube **154** extend colinearly with the longitudinal reference axis L**2**.

Each of the helical vanes **152** extends between an interior surface **162** of the exterior body **150** and an exterior surface **164** of the centralizer tube **154** at least partially along a length of the exterior body **150**. This arrangement of helical vanes **152** relative to the interior surface **162** of the exterior body **150** and the exterior surface **164** of the centralizer tube **154** defines a plurality of helical flow passages **166**. As can be seen, the helical flow passages **166** extend between the exterior body **150**, the centralizer tube **154** and adjacent ones of the helical vanes **152**. The divergent (i.e., tapered) conical profile of the exterior body **150** and the centralizer tube **154** having a generally cylindrical profile (i.e., inside diameter D**1**, shown in FIG. **7**) results in the portions of the helical flow passages **166** within the conically divergent section **160** of the exterior body **150** (i.e., the divergent portion of the helical flow passages **166**) being tapered from a cross-sectional area adjacent to the upstream end portion **158**A of the conically divergent section **158** that is smaller than the cross-sectional area adjacent to the downstream end portion **158**B of the conically divergent section **158**. The convergent (i.e., tapered) conical profile of the exterior body **150** and the centralizer tube **154** having a generally uniform outside diameter (e.g., via cylindrical cross-sectional profile) results in the portions of the helical flow passages **166** within the conically convergent section **160** of the exterior body **150** (i.e., the convergent portion of the helical flow passages **166**) being tapered from a cross-sectional area adjacent to the upstream end portion **160**A of the conically convergent section **160** that is larger than the cross-sectional area adjacent to the downstream end portion **160**B of the conically convergent section **160**. The convergent portions of the helical flow passages **166** are in fluid communication with and extend from the divergent portions of the helical flow passages **166** such that each of the helical flow passages **166** is contiguous along a length thereof. Accordingly, operation of the pump **101** causes material being pump through the helical flow passages **166** to accelerate and to be transform from a laminar flow profile at the material flow inlet **137** of the exterior body **150** to a rotational flow profile at a material flow outlet **139** of the exterior body **150**. For a given profile of the helical vanes **152**, the profile of the interior surface of the exterior body **150** along its length and the profile of the exterior surface of the centralizer tube **154** along its length jointly define the profile of the helical flow passages **166**.

In one or more embodiments, each of the helical flow passages **166** can extend along an entire length of the centralizer tube **154**. In one or more embodiments, an inside diameter of the centralizer tube **154** can be uniform over an entire length of thereof. In one or more embodiments, the centralizer tube can have a cylindrical cross-sectional profile. In one or more embodiments, a maximum inside diameter of the exterior body **150** can be at least 4 times the inside diameter of the centralizer tube **154** and a minimum inside diameter of the exterior body is approximately the same as the inside diameter of the centralizer tube. In one or more embodiments, an inlet of each of the helical flow passages **166** and an inlet of the centralizer tube **154** can all lie in a common plane. In one or more embodiments, the material inlet body **137**, the material outlet body **139** and the centralizer tube **154** can all have nominally the same inside diameter. In one or more embodiments, the material inlet body **137** can have an inside diameter substantially larger than that the of the material outlet body **139**. In one or more embodiments, at least about 80% of a length of the centralizer tube **154** can reside within a portion of the interior space **156** of the exterior body **150** defined by the conically convergent section **160** thereof.

Each of the helical vanes **152** can be fully or partially attached (i.e., e.g., along one or more edge portions thereof) to the exterior body **150**, to the centralizer tube **154** or a combination thereof. In one or more embodiments, the material pressurizer **105** can be formed in a one-piece manner using any suitable fabrication technique (e.g., molding, casting, machining, 3-D printing or the like) and from any suitable material (e.g., metallic material, polymeric material, ceramic material or the like). In one or more other embodiments, one or more components of the material pressurizer **105** can be manufactured as discrete components and subsequently attached to each other by means such as, for example, welding, ultrasonic bonding, adhesive, or the like.

The material flow inlet **137** and the material flow outlet **139** of the exterior body **150** are adapted for enabling attachment of a non-rotating material flow conduit thereto. It is disclosed herein that any suitable means for enabling such attachment of a non-rotating material flow conduit thereto may be used. For example, in one of more embodiments, a rotational-to-static connector **141** (e.g., a commercially-available or custom-fabricated swivel connector) can be attached to the material flow inlet **137** and to material flow outlet **139** for marrying the material flow inlet **137** and the material flow outlet **139** to stationary sections of the material flow conduit. In one or more embodiments, a lubrication system can be integrated into or attached to the rotational-to-static connector to provide intermittent or constant lubrication. Additionally, a thrust washer or other type of device can be added to mitigate or eliminate end-play and/or thermal expansion that may occur between the material pressurizer **105** and the connected non-rotating material flow conduits.

The region of the interior space **156** of the material pressurizer **105** that resides between the downstream end portion **154**B of the centralizer tube **154** and the material flow outlet **139** is a flow mixer section **161** (i.e., convergence of the helical vanes **152**, the central passage **155** of the centralizer tube **154** and tapered region of the exterior body **150** beyond the downstream end portion **154**B of the centralizer tube **154**). Each of the helical flow passages **166** and a central passage **155** of the centralizer tube **154** terminate at the flow mixer section **161**, whereby material from therefrom flows into the flow mixer section **161**. The flow mixer section **161** enhances cyclonic flow efficiency by focusing and centralizing flows toward the longitudinal reference axis L**2**. In preferred embodiments, the focal point of the cyclonic flow of the flowable material is located prior to the material flow outlet **139**. Accordingly, in view of the disclosures made herein, a person of ordinary skill in the art will understand that the duration of strength of the cyclonic flow downstream of the pump **101** is defined by dimensional and structural attributes of the helical flow passages **166**, the centralizer tube **154** and the flow mixer section **161**.

In preferred embodiments, the inertial device **149** comprises a harmonic balancer attached to the exterior body **150** at a longitudinal location at or adjacent to the terminal end of the centralizer tube **154**. Advantageously, this location of the harmonic balancer addresses vibration induced by convergence of the flow from the helical flow passages **166** and the central passage **155** of the centralizer tube **154**.

In one or more embodiments of the disclosures herein, as shown in FIG. **11**, a plurality of helical projections **168** (e.g., ribs, ridges, shoulders or the like) can be provided on the interior surface **167** defining the central passage **155** of the centralizer tube **154**. The helical projections **168** have a height such that the helical projections **168** define a reduced diameter D**2** relative to the inside diameter D**1** of the central passage **155** of the centralizer tube **154**. Preferably, the reduced diameter D**2** is the same as or approximately the same as an inside diameter of the material flow inlet **137** and/or the material flow outlet **139** of the material pressurizer **105**. In one or more embodiments, the helical projections **168** can extend along an entire length of the centralizer tube **154** or can extend partially along an entire length of the centralizer tube **154** (e.g., extending from the inlet of the centralizer tube **154** to an intermedial location thereof, extending from an intermedial location to the centralizer tube **154** to an outlet of the centralizer tube **154** or any segment therebetween).

In one or more embodiments, a pitch of the helical projections **168** can be the same as a pitch of each of the helical vanes **152**. It is disclosed herein the pitch of the helical projections **168** and/or the pitch of each of the helical passages **166** can be constant (i.e., uniform) over the length of thereof or variable over the length thereof. In preferred embodiments, the variable pitch has a looser pitch adjacent to an inlet of the passage and a tighter pitch adjacent an outlet of the passage.

Discussed now are various advantageous aspects of rotary in-line pumps in accordance with embodiments of the disclosures made herein. One such advantageous aspect is that the incorporation of the centralizer tube and resulting helical flow passages provide for cyclonic flow. Such cyclonic flow is characterized by a "top end" or head that is generated by the upstream end portion of the material pressurizer **105** and by omnidirectional flow (i.e., generally equal flow in all directions perpendicular to the axis of rotation). Each of the helical flow passages then uses the imparted energy (i.e., energy from rotational motion of the material pressurizer **105**) and velocity of the material flow to generate several stream vanes of material flow (i.e., helical flow streams) that unite with each other in the flow mixer section **161** and with the material flow of a centralized flow stream (i.e., flow of the centralizer tube **154**). These material flows are then focused by the flow mixer section **161** to the centerline of the material pressurizer **105** (i.e., the longitudinal reference axis L**2**), thereby forming the "tail end" of the cyclonic flow. Beneficially, the flow mixer section **161** further enhances cyclonic flow and distributes an even (i.e., balanced) cyclonic flow profile about the centerline of the material pressurizer **105**. Advantageously, inner sidewall conditions of material flow conduit (e.g., pipeline) downstream of the pump **101** has a negligible effect on the cyclonic flow. Although there is a great deal of energy loss from a fluid going through certain disruptive material flow attributes of material flow conduits (e.g., a valve, fitting, or turbulence created going from passing fluid from one pipe size to another), cyclonic flow mitigates energy loss from these disruptive material flow attributes of material flow conduits by providing for concentration of material flow along the centerline of material flow conduit downstream of the pump **101** thereby reducing sidewall drag and flow resistance.

Another advantageous aspect of rotary in-line pumps in accordance with one or more embodiments of the disclosures made herein is providing for "soft reverse flow". With such soft reverse flow, if there is ever a back flow surge in a system comprising one or more rotary in-line pumps in accordance with one or more embodiments of the disclosures made herein, each of the one or more rotary in-line pumps serves to beneficially reduce the backflow (i.e., flow in the upstream direction). More specifically, in a reverse flow scenario, flowable material enters the helical flow passages from the flow mixer and then dead heads into the 'funnel' of the conically divergent section **158** of the material pressurizer **105**, which creates a controlled flow blockage (i.e., controlled funnel flow). In this regard, soft reverse flow is enabled by inclusion of helical flow passages defined between the exterior body, the centralizer tube and adjacent helical vanes. Such soft reverse flow beneficially does not fully inhibit backflow, which would create a shock wave that is harmful to the structures of the material flow conduit, and to the rotary in-line pump(s).

Still another advantageous aspect of rotary in-line pumps in accordance with embodiments of the disclosures made herein is that they are fully "piggable", as required by the certified in accordance the American Petroleum Institute API-570 inspection process. The oil and petroleum industry require components of pipeline structures to be piggable, which is a process that includes but is not limited to cleaning and inspection of the pipeline interior by deploying a "pigging device" that travels within the pipeline. To this end, rotary in-line pumps in accordance with embodiments of the disclosures made herein permit the pigging device to travel non-obtrusively therethrough regardless of the types of sections that the pipeline includes (e.g., straight line, short radius elbows, long radius elbows, 'Y' fittings, laterals, ellipse, and semi-ellipse cross sections of the pipeline).

The pigging device has an elongated body with a perimeter seal at each of its ends. The perimeter seals have a size whereby they maintain engagement with an inside diameter of a material flow conduit (e.g., pipeline) to support a pressure drop across the length of the pigging device. It is this pressure drop that serves to propel the pigging device along then length of the material flow conduit. This being the case, rotary in-line pumps in accordance with embodiments of the disclosures made herein are configured to maintain engagement between at least one of the perimeter seals and the inside diameter of a material flow conduit and/or rotary in-line pump. More specifically, the length of the centralizer tube of a rotary in-line pump in accordance with embodiments of the disclosures made herein has a length that provides for such seal with the pigging device as it enters and leaves the rotary in-line pumps. As the pigging device passes through the rotary in-line pump, at least one of the perimeter seals is either within portion of the material flow conduit upstream or downstream of the rotary in-line pump or is within the centralizer tube. In some embodiments, the flow inlet structure and/or flow outlet structure can be configured to provide for such seal with the pigging device as it enters and/or leaves the rotary in-line pump.

Although the invention has been described with reference to several exemplary embodiments, it is understood that the words that have been used are words of description and illustration, rather than words of limitation. Changes may be made within the purview of the appended claims, as presently stated and as amended, without departing from the scope and spirit of the invention in all its aspects. Although the invention has been described with reference to particular means, materials and embodiments, the invention is not intended to be limited to the particulars disclosed; rather, the invention extends to all functionally equivalent technologies, structures, methods and uses such as are within the scope of the appended claims.

What is claimed is:

1. A pump, comprising:

a material pressurizer including an exterior body, a centralizer tube, and a plurality of helical vanes each extending between the exterior body and the centralizer tube at least partially along a length of the exterior body to jointly define a plurality of helical flow passages, wherein the exterior body includes a divergent section and a convergent section extending from the divergent section, wherein a first portion of the centralizer tube extends at least partially along a length of the divergent section and a second portion of the centralizer tube extends at least partially along a length of the convergent section, wherein the divergent section and the convergent section are attached to each other such that a maximum diameter of the exterior body is located between opposing end portions thereof, wherein a centerline longitudinal axis of the divergent section extends colinearly with a centerline longitudinal axis of the convergent section, and wherein the divergent section and the convergent section are conically-shaped.

2. The pump of claim **1** wherein a length of the centralizer body is less than a combined length of the divergent section and the convergent section.

3. A pump, comprising:

a material pressurizer including an exterior body, a centralizer tube, and a plurality of helical vanes each extending between the exterior body and the centralizer tube at least partially along a length of the exterior body to jointly define a plurality of helical flow passages, wherein the exterior body includes a divergent section and a convergent section extending from the divergent section and wherein a first portion of the centralizer tube extends at least partially along a length of the divergent section and a second portion of the centralizer tube extends at least partially along a length of the convergent section; and

a plurality of mounting units each engaged with a respective portion of the material pressurizer to enable rotation of the material pressurizer about a longitudinal reference axis, wherein a longitudinal centerline axis of the exterior body and a longitudinal centerline axis of the centralizer tube extend colinearly with the longitudinal reference axis.

4. The pump of claim **3** wherein;

a first one of the mounting units is engaged with at least one of the divergent section and a material inlet body of the exterior body; and

a second one of the mounting units is engaged with at least one of the convergent section and a material outlet body of the exterior body.

5. The pump of claim **3** wherein:

the exterior body includes a material inlet body at an upstream end portion thereof and a material outlet body at a downstream end portion thereof; and

a longitudinal centerline axis of the material inlet body and a longitudinal centerline axis of the material outlet body each extend colinearly with a centerline longitudinal axis of the centralizer body.

6. The pump of claim **5** wherein the centralizer tube and at least one of the material inlet body and the material outlet body each have a common inside diameter that is uniform over an entire length of thereof.

7. The pump of claim **5** wherein each of the helical flow passages extends along an entire length of the centralizer tube.

8. The pump of claim **7** wherein:

the divergent section and the convergent section are attached to each other such that a maximum diameter of the exterior body is located between opposing end portions thereof, and

at least one of the divergent section and the convergent section is conically-shaped.

9. The pump of claim **8** wherein the centralizer tube and at least one of the material inlet body and the material outlet body each have a common inside diameter that is uniform over an entire length of thereof.

10. The pump of claim **7** wherein at least one of:

an inside diameter of the centralizer tube is uniform over an entire length of thereof; and

a length of the centralizer body is less than a combined length of the divergent section and the convergent section.

11. The pump of claim **5** wherein at least one of:

an inside diameter of the centralizer tube is uniform over an entire length of thereof; and

a length of the centralizer body is less than a combined length of the divergent section and the convergent section.

12. A pump, comprising:

a material pressurizer including an exterior body, a centralizer tube, and a plurality of helical vanes each extending between the exterior body and the centralizer tube at least partially along a length of the exterior body to jointly define a plurality of helical flow passages, wherein the exterior body includes a divergent section and a convergent section extending from the divergent section, wherein a first portion of the centralizer tube extends at least partially along a length of the divergent section and a second portion of the centralizer tube extends at least partially along a length of the convergent section, and wherein each of the helical flow passages extends along an entire length of the centralizer tube.

13. The pump of claim **12** wherein an inside diameter of the centralizer tube is uniform over an entire length of thereof.

14. The pump of claim **12** wherein a length of the centralizer body is less than a combined length of the divergent section and the convergent section.

15. The pump of claim **14** wherein an inside diameter of the centralizer tube is uniform over an entire length of thereof.

16. The pump of claim **12** wherein:

the divergent section and the convergent section are attached to each other such that a maximum diameter of the exterior body is located between opposing end portions thereof, and

at least one of the divergent section and the convergent section is conically-shaped.

17. The pump of claim **16** wherein at least one of:

an inside diameter of the centralizer tube is uniform over an entire length of thereof; and

an inside diameter of the centralizer tube is uniform over an entire length of thereof.

18. A pump, comprising:

a material pressurizer including an exterior body, a centralizer tube, and a plurality of helical vanes each extending between the exterior body and the centralizer tube at least partially along a length of the exterior body to jointly define a plurality of helical flow passages, wherein the exterior body includes a divergent section and a convergent section extending from the divergent section, wherein a first portion of the centralizer tube extends at least partially along a length of the divergent section and a second portion of the centralizer tube extends at least partially along a length of the convergent section, wherein an inlet of each of the helical flow passages and an inlet of the centralizer tube all lie in a common plane, and wherein at least one of: an inside diameter of the centralizer tube is uniform over an entire length of thereof; and a length of the centralizer body is less than a combined length of the divergent section and the convergent section.

* * * * *